

We built a lab instrument without CAD software—artificial intelligence designed every part

An open-source, 3D-printed powder doser as a case study in AI-assisted hardware design

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In one sentence: We used generative AI to design a working laboratory machine from start to finish, and the most important result is not the machine itself—it is what we learned about letting AI do the engineering.

Background

Modern research labs increasingly run experiments automatically, with robots mixing chemicals and machine-learning software deciding what to try next. These “self-driving labs” are good at handling liquids but still struggle with powders, which most labs weigh out by hand. Machines that dose powders automatically do exist, but they cost tens of thousands of dollars, putting them out of reach for most academic groups.

At the same time, AI tools that can write code and even generate 3D shapes are improving quickly. We wanted to know a practical question: can today’s AI actually design a real, physical, working instrument—not just a shape on a screen—if a small team guides it? To find out, we set ourselves a strict rule and a real goal.

Design solution

The goal was a low-cost powder doser that any lab could 3D-print and build. Our design uses a turning screw (an auger) to push powder out a nozzle, a small tapping and vibrating mechanism to keep the powder flowing, a tilting base so the powder always lands in the same spot, and a precise balance that weighs the powder as it falls and tells the machine when to stop. A \$6 microcontroller runs everything. One complete unit can be built for under \$200, roughly one hundred times cheaper than commercial systems.

For this conference the machine is mainly an example. The point we want to make is not that it dispenses powder well, but that it was designed in an unusual way.

Generative AI

Here is the strict rule we followed: **no traditional CAD software was used at any point.** Engineers normally design parts by hand in programs like SolidWorks or Fusion 360. We never opened them. Instead, every part was made by AI.

We worked in two ways. Most parts were built by AI “coding agents” that write the design as computer code; the code is turned into a 3D model and a picture automatically, and our team reviewed each result and asked for fixes. Late in the project we also used Zoo Design Studio and its AI agent, Zookeeper, which we drove almost entirely by typing instructions in plain language rather than by drawing.

Two lessons stood out. First, **asking the AI to design the whole machine at once did not work**—the parts overlapped, floated in mid-air, or left no room for the motors. But **asking for one part at a time, with clear instructions about how it connects to its neighbors, worked well.** Second, **the AI needed constant checking.** It would sometimes quietly change a part we had already finished, so every new version had to be re-inspected. Automated checks and careful human review caught most problems.

We were honest about the trade-offs. An experienced engineer using normal CAD software would have been

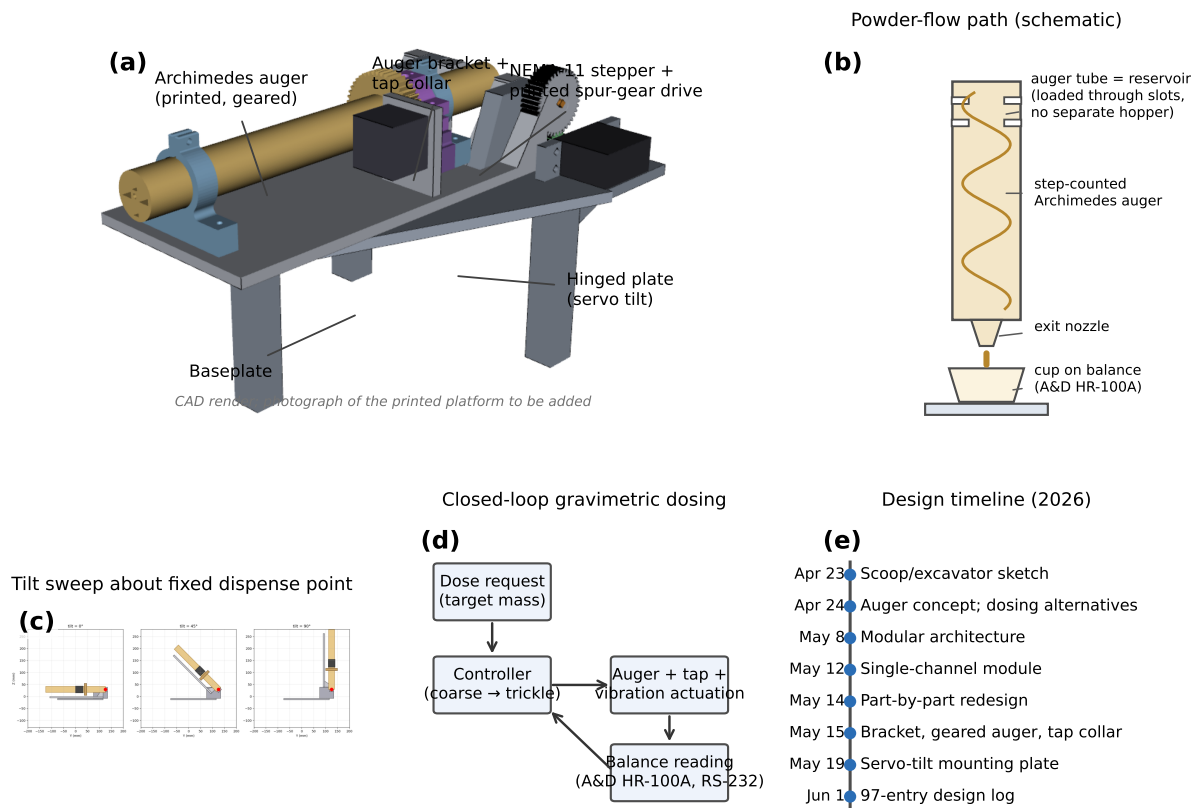


Figure 1: The powder doser. Humans chose the overall design and reviewed, printed, and tested every part; AI tools created the 3D models. No traditional CAD software was used.

faster and produced cleaner parts. What the AI approach gave us instead was a complete, written record of every decision, every prompt, and every mistake. That record makes the whole project reproducible and easy for others to learn from—something a normal CAD session leaves behind only in a designer’s head.

Conclusion and future work

We designed and built a real, working laboratory instrument entirely with generative AI, under human direction, and without ever touching conventional CAD software. The honest, complete history of the project—including its failures—is public, so it doubles as a case study of what AI can and cannot yet do in hardware design.

Next, we are extending the design into a multi-powder version, adding automatic calibration, and continuing to test new AI design tools. Every design file, the firmware, and the full log of AI prompts and responses are freely available at <https://github.com/vertical-cloud-lab/powder-doser>.